Ian Helfrich, Ph.D.

APPLIED ECONOMIST · QUANTITATIVE RESEARCH DESIGNER · EDUCATOR & SYSTEMS BUILDER

ianthelfrich@gmail.com · ihelfrich.github.io · github.com/ihelfrich · LinkedIn

PROFESSIONAL PROFILE

Applied economist, quantitative research designer, educator, and systems builder. I turn consequential questions whose data and estimands are not yet cleanly separated into auditable research designs, reproducible analyses, and explanations suited to researchers, executives, professionals, and students. Research and teaching are complementary parts of the same practice: establish what can responsibly be learned, then build the curricula, software, diagnostics, and visual explanations that make the reasoning transferable.

PROFESSIONAL AND ADVISORY EXPERIENCE

Independent Economist and Quantitative Researcher | Self-directed and collaborative applied research — 2024-present

- Lead work from substantive problem definition through data construction, estimand design, model diagnostics, interpretation, visualization, and manuscript or decision-facing communication.
- Build reproducible evidence systems that preserve provenance, expose assumptions and failure modes, and distinguish a completed computation from a supported inference.

Independent Economic Advisor | Industrial-sector diagnostics — 2026-present

- Developed an industrial-sector diagnostic methodology covering indicator selection, analytical sequencing, composite-index construction, and international benchmarking.
- Structure complex advisory questions as transparent sequences of evidence, judgment, and decision rather than opaque composite scores.

Independent Tutor and Quantitative Research-Design Coach | Wyzant and direct practice — 2021-present

- Provide individualized instruction in economics, statistics, econometrics, finance, strategy, quantitative marketing, research design, Python, SQL, Stata, and GIS from introductory through doctoral, professional, and executive levels.
- Help researchers and professionals define the decision, estimand, data requirements, credible comparisons, diagnostics, robustness checks, and communication boundaries for quantitative work.
- More than 1,035 hours through Wyzant as of August 2026, with a 5.0/5 public rating; nearly 1,000 additional direct tutoring and executive-level research-design coaching hours outside the platform.

RESEARCH

CURRENT WORKING PAPERS

Helfrich, Ian. "The Rural Mobilization Gap in a U.S. Place-Based Tax Credit."

Sole author and originator · Working paper, August 2026 · With contributions from Katia Antunes and Elizaveta Gonchar

- Builds a project-level panel from 19,907 QLICI transactions covering 8,024 projects with origination years 2001-2022.
- Uses a noncausal fixed-effects and Gelbach decomposition of observed rural-urban leverage differences. CDE identity contributes -0.185, or 86% of explained movement in the base decomposition, and -0.1824, or 88.1%, when purpose of investment enters as a fourth block.
- Finds no excess mass around the 20% target in realized cumulative deployment; allocation-agreement compliance is unobserved.

Helfrich, Ian. "Human or Machine? Out-of-Time Validation of Task-Level AI Exposure Ratings."
Sole-authored working paper · August 2026

- Evaluates the out-of-time measurement validity of trained-human and GPT-4 task-level AI-exposure ratings across 19,265 occupation-task records and 923 occupations.
- Separates measurement validity from employment-effect claims and rejects a mechanically circular validation outcome.

Helfrich, Ian, and Shane Vardanyan. "AI and the Entry-Level Labor Market."
Working paper in active development · No results claimed

Designs continuous-exposure event-study and triple-difference analyses using Current Population Survey microdata.

PREPRINTS AND LONG-FORM WORK

Gonchar, Elizaveta, and Ian Helfrich. "Trade in the Spotlight: Enhancing Gravity Model Predictions with Nightlights and Population-Weighted Distance Measures."
Preprint · SSRN 5202676 · 2025

Constructs annual population- and nightlight-weighted distance measures and evaluates them in gravity models of interstate and international trade.

Helfrich, Ian. "A Unifying Framework for Economic Equilibrium and Optimal Transport in Infinite-Dimensional Hilbert Spaces."
Working paper · SSRN 4772016 · 2024

Helfrich, Ian, and Elizaveta Gonchar. "Pictures of Inference."
Book project in progress · Part One chapters 1-10 drafted as of August 14, 2026

Visual statistics and econometrics paired with a practice workbook; later parts remain in development.

SELECTED RESEARCH IN DEVELOPMENT — STATUS STATED EXPLICITLY

- **Effective Distance** — with Elizaveta Gonchar; in preparation. Planned 2000-2024 bilateral panel and structural-gravity validation; no completed-output or counterfactual-result claim.
- **Ukraine under sustained shock** — with Elizaveta Gonchar; research program in preparation. Planned spatial, labor, and trade measurement; no empirical results claimed.
- **Russian seaborne crude under sanctions** — design and data construction; no empirical results yet.
- **Penumbra** — sole-authored draft on observation under measurement gaps; not yet posted to SSRN.

DISSERTATION AND POLICY CONTRIBUTION

Redefining Trade Dynamics: Incorporating Network Centrality, Spatial Heterogeneity, and Optimal Transport in International Trade.
Doctoral dissertation · Georgia Institute of Technology · 2024

Three essays integrating trade-network analysis, panel and gravity econometrics, satellite-derived spatial measures, and optimal-transport theory.

Securing America's Future: A Framework for Critical Technology Assessment.
National Network for Critical Technology Assessment · 2023 · NSF-sponsored national report

Additional contributor to the semiconductor demonstration; this is project credit, not authorship of the full report.

SELECTED DOCTORAL RESEARCH COLLABORATION

- **Georgia Tech Electrical and Computer Engineering:** applied tensor-decomposition and signal-processing concepts to spatial-economic panel questions; supported quantitative teaching in ECE alongside economics.
- **Georgia Tech public-policy work:** contributed quantitative analysis and visualization to critical-technology assessment and workforce-oriented research; public credit is stated separately above at the verified NNCTA level.

TEACHING AND QUANTITATIVE COACHING

Instructor of Record and Graduate Teaching Assistant | Georgia Institute of Technology — doctoral appointment

- Instructor of record for three semesters: ECON 2100 twice and ECON 2106; provided additional teaching support across economics and electrical and computer engineering.
- Connected economic reasoning, mathematical structure, empirical implementation, and interpretation in lectures, sections, recitations, assessment, and office hours.
- Georgia Tech Economics Graduate Teaching Assistant of the Year, 2023.

Graduate Teaching Fellow | Georgia Tech Center for Teaching and Learning — 2018-19

Member of the inaugural cohort; designed professional-development curriculum and co-led orientation for new and returning teaching assistants and instructors.

Teaching Support | Duke University Global Executive MBA — 2016-17

Supported strategy, managerial economics, and global-markets instruction; connected economic frameworks and quantitative reasoning to executive decisions.

Graduate Teaching Assistant | Indiana University Bloomington — 2015-16

Supported undergraduate economics instruction while completing graduate study in economics.

Online Instructor | Campus / MTI College cohorts — 2022-23

Principles of Microeconomics, Summer and Fall 2022; Principles of Macroeconomics, Spring 2023.

Independent Tutor and Quantitative Research-Design Coach | Wyzant and direct practice — 2021-present

- Teach economics, statistics, econometrics, research design, Python, SQL, Stata, finance, strategy, quantitative marketing, and GIS to students, graduate researchers, and working professionals.
- Provide executive-level coaching on how to formulate, execute, test, and communicate quantitative research while leaving a reproducible path for independent continuation.

CURRICULUM AND OPEN EDUCATIONAL INFRASTRUCTURE

Statistics, Inference, and Econometrics | modular methods library
Reusable explanations, visual models, public-data exercises, regression workflows, and worked problems that separate durable concepts from Excel, Stata, R, Julia, and browser-based implementation.

Microeconomics and Macroeconomics | principles through applied analysis
Structured lessons, live-data dashboards, quantitative exercises, and decision-focused examples spanning individual choice, markets, growth, stabilization, money, and international linkages.

Research Computing and Data Practice | Python · SQL · reproducible workflows
Lessons, diagnostics, code exercises, data provenance, and assessment rehearsal designed to build independent analytical fluency rather than solve a single assignment.

Business, Strategy, and Quantitative Marketing | executive and professional audiences
Decision-centered coaching that connects economic frameworks, finance, empirical design, analytics, and communication to business questions.

Spatial and Public-Data Methods | GIS · Census · FRED
Reusable training sequences connecting geographic representation, source provenance, statistical analysis, visualization, and policy interpretation.

Pictures of Inference | visual statistics and econometrics
Book and practice-workbook project, with Elizaveta Gonchar, for making assumptions, estimands, sampling variation, and causal reasoning visible.

EDUCATION

Ph.D., Economics, Georgia Institute of Technology | Doctoral minor: Quantitative Psychology	2024
M.S., Geographic Information Science and Technology, Georgia Institute of Technology	2022
M.S., Economics, Georgia Institute of Technology	2019
M.A., Economics, Indiana University Bloomington	2018
M.Sc., Specialized Economic Analysis, Barcelona School of Economics | Economics and Public Policy	2015
B.A., Economics, University of North Carolina at Chapel Hill	2014

SELECTED ANALYTICAL AND LEARNING SYSTEMS

NMTC evidence system | public finance · Python/R · Quarto
Pipeline, 204-claim manuscript audit, replication archive, and interactive evidence viewer linking data provenance to displayed claims.

Laborscope | labor-market observatory · Python/SQLite
Longitudinal source-event histories, inspectable rankings, and failure controls that distinguish incomplete retrieval from genuine disappearance.

Sector diagnostic framework | industrial analysis · Python/R/SQL
Reproducible architecture for sequenced indicators, trade and production evidence, composite diagnostics, and international benchmarking.

Browser-Based Econometrics Laboratory | executable quantitative instruction
Role-separated learner and instructor builds, frozen teaching datasets, regression exercises, familiar statistical artifacts, and reproducibility/parity controls.

METHODS AND TECHNICAL PRACTICE

Research design	estimand definition; data fitness; descriptive/causal boundaries; panel and fixed-effects models; Gelbach decomposition; event-study, difference-in-differences, and triple-difference designs; quantile and median regression; clustered and bootstrap inference; measurement validation.
Substantive domains	public finance and place-based policy; international trade and economic networks; labor and technology measurement; geospatial economics; critical-technology and industrial diagnostics.
Evidence systems	data provenance; reproducible pipelines; adversarial diagnostics; versioned research artifacts; decision logs; interactive evidence products; technical reporting and visualization.
Working tools	R; Python; SQL with SQLite and DuckDB; Stata; Julia; Excel; QGIS and ArcGIS Pro; Git; LaTeX, Typst, and Quarto.

HONORS, REVIEW AND LEADERSHIP

Georgia Tech Economics Graduate Teaching Assistant of the Year	2023
Georgia Tech Graduate Student Government Senator of the Year	2020
Referee, *Journal of Economic Theory*	2025

Graduate Student Government, Georgia Institute of Technology | 2018-20
Vice President for Campus Services and graduate senator; coauthored a policy white paper and represented graduate students in cross-campus planning and service discussions.

SELECTED PUBLIC EVIDENCE

Portfolio	ihelfrich.github.io
Research code	github.com/ihelfrich
NMTC project	github.com/ihelfrich/us-nmtc-viewer
Contact	ianthelfrich@gmail.com

This CV states current research status explicitly and omits unresolved client identities, stale personal contact data, and unverified metrics.